

WENDA CHU

wchu@caltech.edu

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ABOUT ME

I am a third-year PhD student at Caltech, advised by Yisong Yue. My research focuses on **generative modeling**, in particular **inference-time posterior sampling algorithms with pretrained diffusion priors**, with application to reward-guided generation and scientific inverse problems. More recently, I have been working on developing more generation-friendly visual tokenizers for latent generative models.

EDUCATION

California Institute of Technology

September 2023 - Present

Ph.D. student, Computing and Mathematical Science

Tsinghua University, Beijing, China

September 2019 - July 2023

Bachelor of Computer Science, Yao Class, IIIS

GPA: 3.91/4.00

PUBLICATIONS

(* FOR EQUAL CONTRIBUTION)

[1] **End-to-End Autoregressive Image Generation with 1D Semantic Tokenizer**

Wenda Chu, Bingliang Zhang, Jiaqi Han, Yizhuo Li, Linjie Yang, Yisong Yue, Qiushan Guo.

ICML 2026 (Spotlight).

[2] **SpeedDiff: Scalable Pixel-Anchored End-to-End Latent Diffusion Model**

Bingliang Zhang, Wenda Chu, Yizhuo Li, Linjie Yang, Yisong Yue, Katherine Bouman, Yang Song, Qiushan Guo.

CVPR 2026.

[3] **Discrete Diffusion Trajectory Alignment via Stepwise Decomposition**

Jiaqi Han*, Austin Wang*, Minkai Xu, Wenda Chu, Meihua Dang, Haotian Ye, Huayu Chen, Yisong Yue, Stefano Ermon.

ICLR 2026.

[4] **How Large Language Models Balance Internal Knowledge with User and Document Assertions**

Shuwei Li, Haoxin Li, Wenda Chu, Yi Fang.

ACL 2026 Findings.

[5] **Split Gibbs Discrete Diffusion Posterior Sampling**

Wenda Chu, Zihui Wu, Yifan Chen, Yang Song, Yisong Yue.

NeurIPS 2025.

[6] **Steering Generative Models with Experimental Data for Protein Fitness Optimization**

Jason Yang*, Wenda Chu*, Daniel Khalil, Raul Astudillo, Bruce Wittmann, Frances H. Arnold, Yisong Yue.

NeurIPS 2025.

[7] **Improving Diffusion Inverse Problem Solving with Decoupled Noise Annealing**

Bingliang Zhang*, Wenda Chu*, Julius Berner, Chenlin Meng, Anima Anandkumar, Yang Song.

CVPR 2025 (Oral).

[8] **InverseBench: Benchmarking Plug-and-Play Diffusion Models for Scientific Inverse Problems**
Hongkai Zheng*, **Wenda Chu***, Bingliang Zhang*, Zihui Wu*, Austin Wang, Berthy Feng, Caifeng Zou, Yu Sun, Nikola Borislavov Kovachki, Zachary E Ross, Katherine Bouman, Yisong Yue.
ICLR 2025 (Spotlight).

[9] **Ensemble Kalman Diffusion Guidance: A Derivative-free Method for Inverse Problems**
Authors: Hongkai Zheng, **Wenda Chu**, Austin Wang, Nikola Kovachki, Ricardo Baptista, Yisong Yue.
TMLR.

[10] **COMMIT: Certifying Robustness of Multi-Sensor Fusion Systems against Semantic Attacks**
Zijian Huang*, **Wenda Chu***, Linyi Li*, Chejian Xu, Bo Li.
AAAI 2025.

[11] **PerAda: Parameter-Efficient and Generalizable Federated Learning Personalization with Guarantees**
Chulin Xie, De-An Huang, **Wenda Chu**, Daguang Xu, Chaowei Xiao, Bo Li, Anima Anandkumar.
CVPR 2024.

[12] **Physically Realizable Natural-Looking Clothing Textures Evade Person Detectors via 3D Modeling**
Zhanhao Hu*, **Wenda Chu***, Xiaopei Zhu, Hui Zhang, Bo Zhang, Xiaolin Hu.
CVPR 2023.

[13] **FOCUS: Fairness via Agent-Awareness for Federated Learning on Heterogeneous Data**
Wenda Chu*, Chulin Xie*, Boxin Wang, Linyi Li, Lang Yin, Han Zhao, Bo Li.
FL@FM Workshop at NeurIPS 2023 (Oral).

[14] **TPC: Transformation-Specific Smoothing for Point Cloud Models**
Wenda Chu, Linyi Li, Bo Li.
ICML 2022.

EXPERIENCE

Research Internship at ByteDance Seed *June-September 2025*

- Research on end-to-end latent diffusion model training with accelerated convergence rate and a generation-friendly latent space.
- Research on training 1D causal vision tokenizer with high compression rate, empowering autoregressive image generation in the latent space.

Part-time Research Internship at Pika Lab *September 2023-February 2024*

- Research on training diffusion models for guided video generation.

Research Assistant at University of Illinois, Urbana Champaign *October 2021-May 2022*

- Trustworthy machine learning, advised by Prof. Bo Li.

HONORS AND AWARDS

- Kortschak Scholarship 2024
- Scientific Innovation Excellence Award - Tsinghua University 2022
- Sports Excellence Award - Tsinghua University 2019, 2020, 2021
- **2nd** place in the 35th Chinese Physics Olympiad (CPhO) 2018